RemindCare

Technical guide for **pregnancy-reminder-bot**, a WhatsApp companion for pregnant mothers with an admin website for health workers.

Version 1.0 · September 2026 · describes the code in this repository as of commit f776a44

RemindCare reminds a mother every day to take her iron tablet (tablet FE). From week 39 it asks whether she has given birth, records the birth, and then reminds her about the four postpartum check-ups for her and her baby (KF/KN). Everything happens in an ordinary WhatsApp chat. Health workers follow progress on a small website.

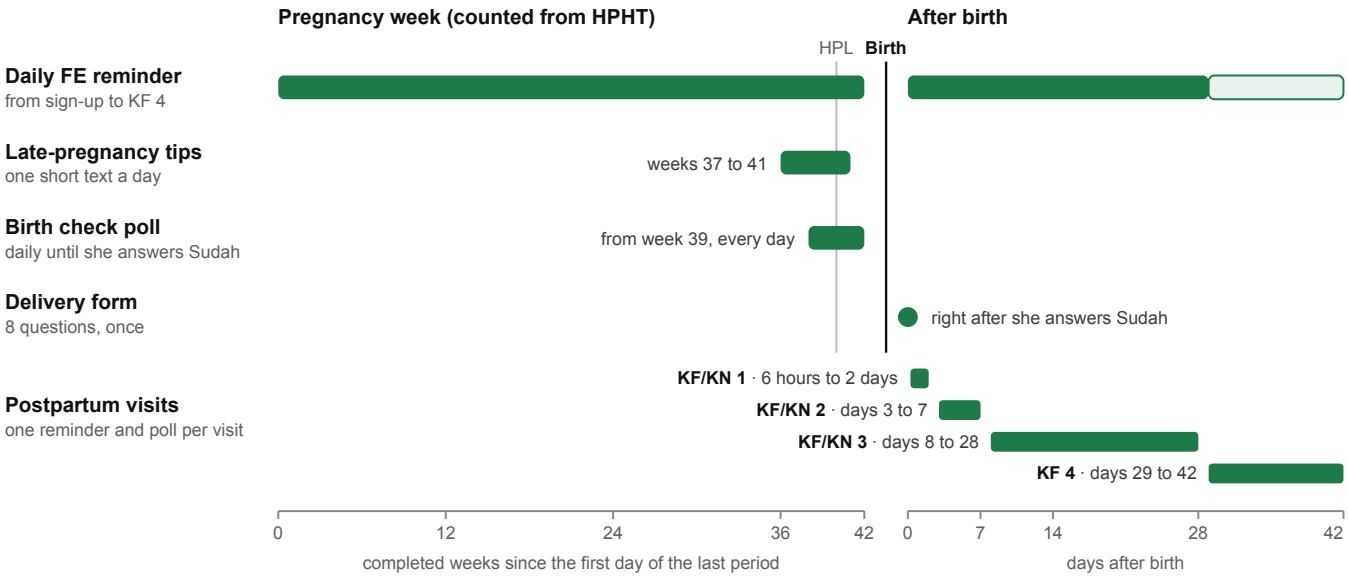


Figure 1. The whole program at a glance. Dark bars are periods when the bot sends messages. HPL (due date) is HPHT plus 280 days. The light end of the FE bar means the reminder keeps running until the mother confirms KF 4 (see known issue 4 on page 8).

How to read this guide

Program staff (midwives, supervisors, reviewers): pages 3, 4, 5 and 7 explain what a mother experiences and what you can control from the website.

Developers and operators: pages 2, 6 and 8 cover how the bot is built, how it avoids being flagged by WhatsApp, where the data lives, and how to run it on a server.

Bot messages are quoted in Indonesian exactly as the bot sends them, with an English translation underneath.

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At a glance

Runtime	Node.js 18 or newer, one process (`index.js`)
WhatsApp	whatsapp-web.js 1.34 driving a hidden Chrome browser. The bot is a linked device on a normal number, not the WhatsApp Business API.
Storage	One SQLite file, `data/remindcare.db`, in WAL mode
Clock	Asia/Jakarta (WIB). The scheduler checks every 30 seconds.
Admin	Express website on `127.0.0.1:3030`, login required
Hosting	systemd service plus a daily backup timer (`deploy/`)
Tests	`npm test`: 8 test files, 61 checks, all passing on 29 September 2026

One process, one gate for every message

RemindCare is a single Node.js process. It holds a WhatsApp session through a hidden Chrome browser, answers mothers, checks every 30 seconds who is due for a reminder, and serves the admin website. Every message it sends passes through one gate: the send guard.

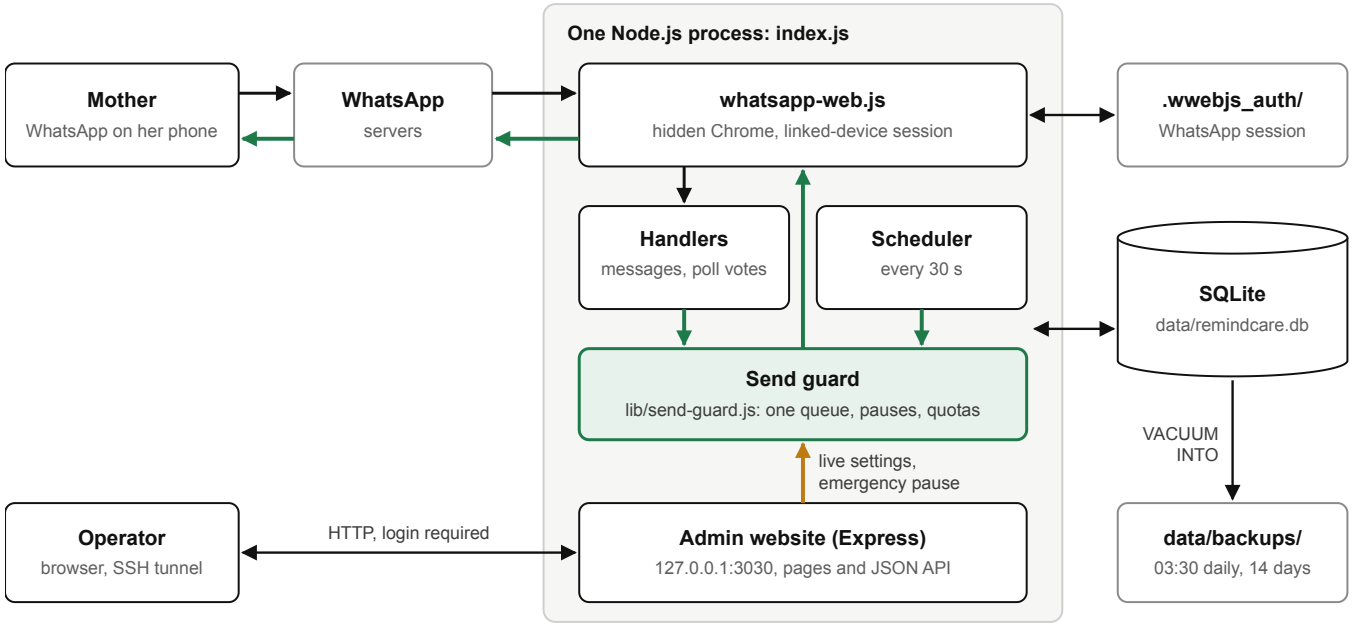


Figure 2. System overview. **Green**: messages to mothers, always through the send guard. **Black**: messages from mothers and data access. **Amber**: the operator changing how sending behaves. Every part of the process reads and writes the same SQLite file.

Startup order

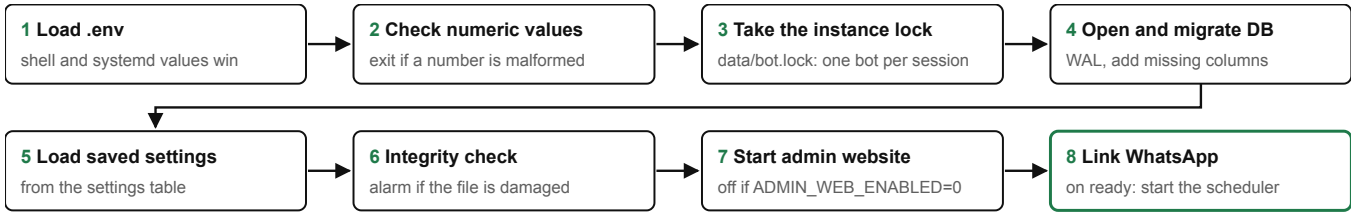


Figure 3. What happens on `npm start`. The scheduler only starts once WhatsApp reports ready, so nothing is sent while the session is still linking. On SIGTERM it closes Chrome and the database, then releases the lock.

Files that matter

File	What it does
index.js	Settings, handlers, scheduler rules, database, admin pages (about 7,300 lines)
lib/send-guard.js	Outbound queue: pauses, quotas, send window, warm-up, circuit breaker
lib/admin-settings-page.js, lib/env-file.js	Settings page HTML; `.env` loader that keeps values already set
scripts/backup-db.js	Daily copy, integrity check, 14-day rotation (`npm run backup`)
scripts/rotate-admin-password.js	New admin password into `data/admin_web_password.txt`, never printed
scripts/reset-seed-db.js	Deletes the database, loads four demo mothers. Never run on a server.
deploy/, tests/	systemd units and INSTALL.md; 8 test files run by `npm test`

Alarms. The bot raises an alarm when the WhatsApp session drops or fails to authenticate, the circuit breaker trips, the database fails its integrity check, or the daily sign-up limit is reached. Alarms go to the log and, if `ALERT_WEBHOOK_URL` is set, are posted there as JSON, at most once per 30 minutes per alarm. That variable is not in `.env.example` yet.

Nine questions, then the bot runs on its own

A mother joins by sending **start** to the bot's number and answering nine short questions. After that, reminders arrive at the time she chose. Her record is always in one of four states.

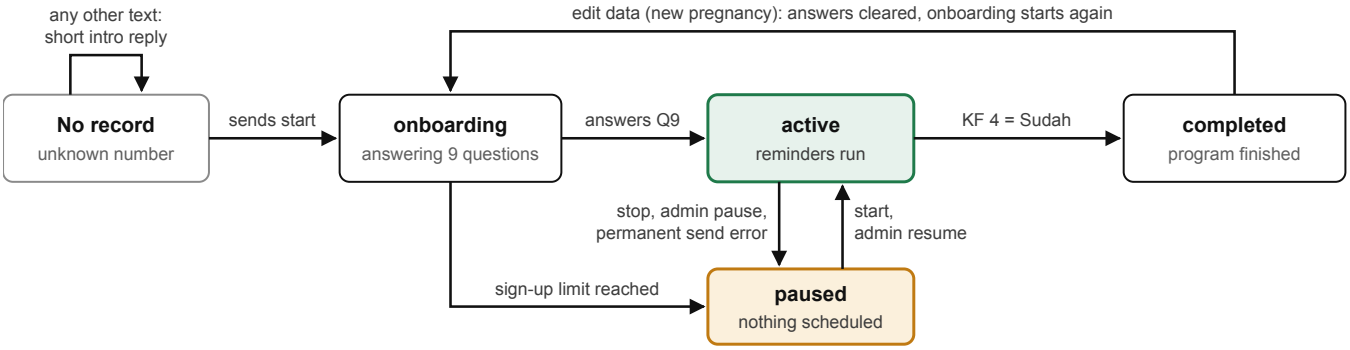


Figure 4. The `status` column of a mother's record. Sending **delete** twice within 5 minutes erases the record and all its logs, which returns the number to "No record". Answering *tidak* to question 8 makes her active with reminders switched off. A record also completes after week 42 if she confirmed the birth but never finished the delivery form.

The nine questions

1 **Name** free text	2 **Age** free text	3 **Which pregnancy** free text
4 **HPHT** past date, dd-mm-yyyy	5 **Routine medicine** ya / tidak	6 **Drinks tea** ya / tidak
7 **Who reminds her** free text	8 **Wants reminders** ya / tidak, consent	9 **Reminder time** 17:00, 17.00 or 1700

- A wrong format gets a short hint and the same question again.
- HPHT (first day of the last period) sets the due date, HPL = HPHT + 280 days, and every pregnancy week in this guide.
- First reminder: today if her time has not passed yet, else tomorrow.
- At most 20 new sign-ups a day (`onboarding_daily_limit`).

WhatsApp, as the bot sends it

Pertanyaan 4 dari 9
HPHT (Hari Pertama Haid Terakhir) kapan? Format tanggal-bulan-tahun, contoh: 31-01-2024

Question 4 of 9. When did your last period start? Day-month-year.

31/13/2025

Format HPHT belum sesuai. Contoh: 31-01-2024.

That date is not valid. Example: 31-01-2024.

10-01-2026

What a mother can type

Message	Effect
start, mulai	Reminders on (asks for a time if none)
stop, berhenti	Pause every reminder
ubah jam 17:00	New reminder time, reminders on
info	Her data, HPL, birth and visit status
menu, help	List of commands
sudah, belum	Answer the open question by text

Message	Effect
cek persalinan	Birth question now, any week
edit data	Clear all, restart onboarding *
edit persalinan	Redo the delivery form *
delete, hapus	Erase her record and logs *
batal	Cancel * or stop the delivery form
koreksi kf2 sudah	Correct one visit answer

* Needs the same message twice within 5 minutes. Group chats and status updates are ignored. Voice notes and pictures get a request to type. More than 20 messages in a minute triggers a 2-minute cooldown. With the allowlist on, unknown numbers are told *Nomor ini belum diizinkan* (this number is not allowed yet).

One reminder a day, on time or not at all

Each mother gets at most one iron-tablet reminder a day, at the time she picked. If the bot misses that moment by more than 30 minutes, for example after a restart, the reminder is dropped for the day instead of arriving late. This keeps a restart from sending a burst of old reminders to everyone at once.

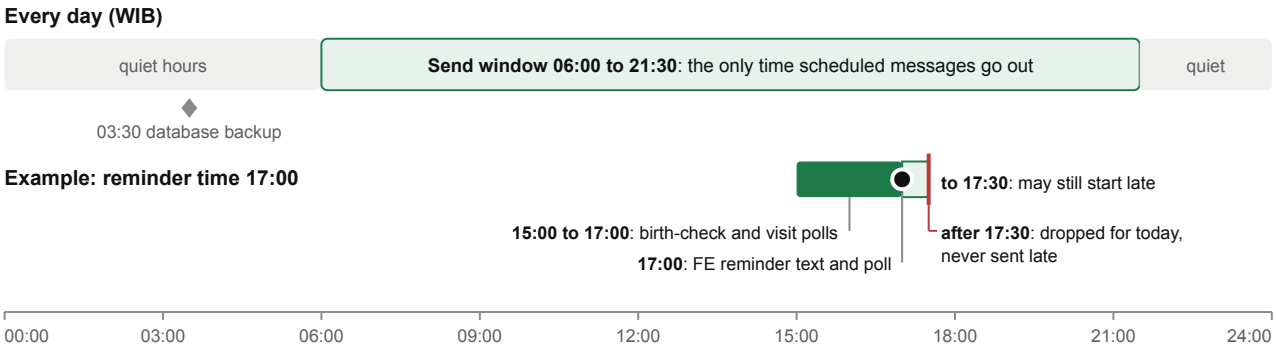


Figure 5. The two windows every mother has. The 2-hour window before her time is used for the birth check (from week 39) and postpartum visit polls. Her reminder may start up to 30 minutes late (`reminder_stale_after_minutes`); after that it is skipped. A time outside 06:00 to 21:30 never fires (known issue 6).

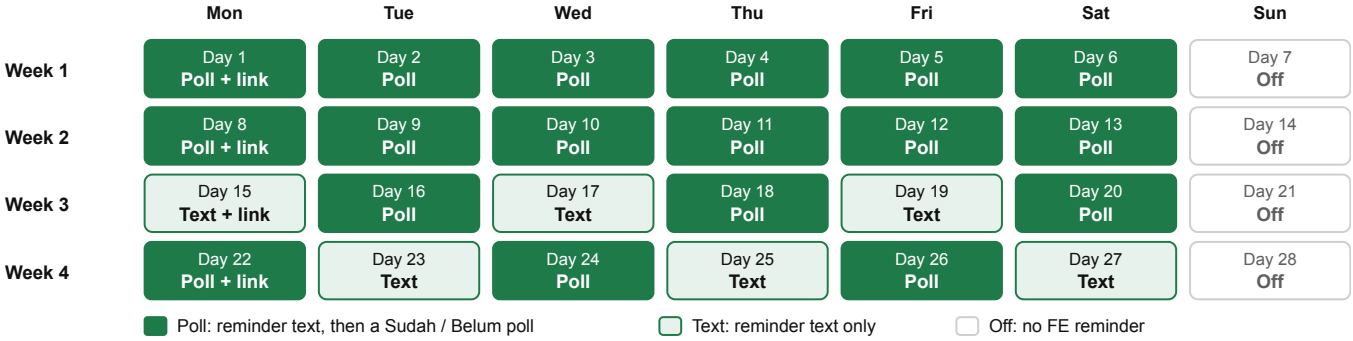


Figure 6. What a mother receives in her first four weeks, if she signs up on a Monday before her reminder time. After day 14 (`poll_days_limit`) the poll goes out every other day and the other days carry only the text (which currently arrives twice, known issue 3). Sundays have no FE reminder (`reminder_skip_weekday` = 7). Mondays add the article link.

Rules the scheduler follows

- It only looks at mothers who are **active**, have reminders on and a time set, and are not blocked (and are allowed, when the allowlist is on).
- Wording is picked per mother per day from 10 reminder texts and 7 poll questions, so the same sentence does not go to everyone.
- Weeks 37 to 41: a short tip for that week goes out just before the reminder. The week 40 tip includes her HPL.
- From week 39 the reminder text is left out and only the poll goes, because the birth-check poll already came that day.
- No more than 3 scheduled sends per mother per day; a text with its poll counts as one (`max_messages_per_user_per_day`).
- Each answer type is counted at most twice a day. A failed send is retried later (page 6). If WhatsApp reports the number as unregistered or blocked, she is paused.

WhatsApp, as the bot sends it (emoji left out)

Selamat sore, [nama].
Saatnya tablet FE hari ini.
Good afternoon, [name]. Time for today's FE tablet.

Sudah minum tablet FE hari ini?

- (●) Sudah
- ( ) Belum

Have you taken today's FE tablet? (she taps Sudah)

Terima kasih. Semoga sehat selalu.
Thank you. Stay healthy.

From "have you given birth?" to the last check-up

From week 39 the bot asks once a day whether the mother has given birth. When she answers *Sudah* (yes), it collects eight details about the birth, then reminds her about four check-ups for her and her baby over the next six weeks.

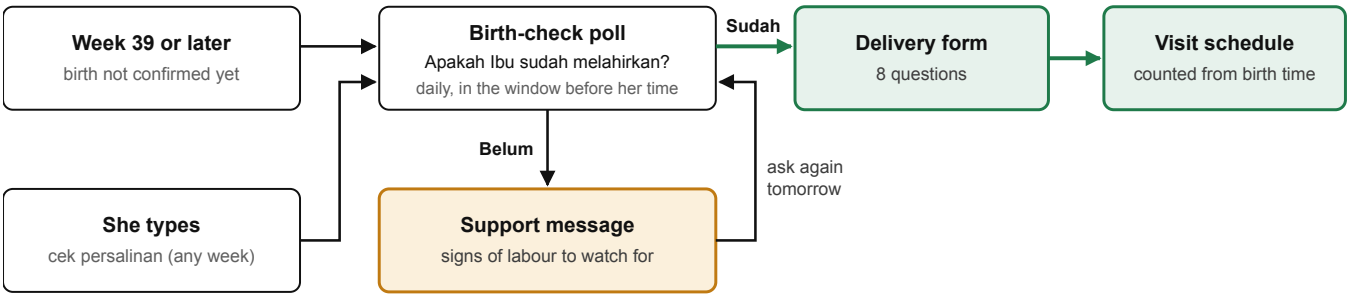


Figure 7. The birth check. A *Belum* (not yet) answer gets a supportive message listing labour signs such as regular contractions or waters breaking, and the poll comes back the next day.

The delivery form

1. **Date of birth**: not before HPHT, not in the future
2. **Time of birth**: 24-hour; date plus time not in the future
3. **Place**: home, puskesmas, clinic, hospital
4. **Birth attendant**: midwife, doctor, other
5. **Complications**: ya / tidak
6. **Baby's sex**
7. **Birth weight** in grams, for example 3200
8. **Mother's complaints** now, or *tidak ada*

After the form

- The answers are saved, any old visit schedule is cleared, and she gets one message about why the postpartum weeks matter.
- **batal** stops the form and keeps the answers so far. **edit persalinan** starts it again.
- From here on the bot tracks four visits, shown below. FE reminders continue.

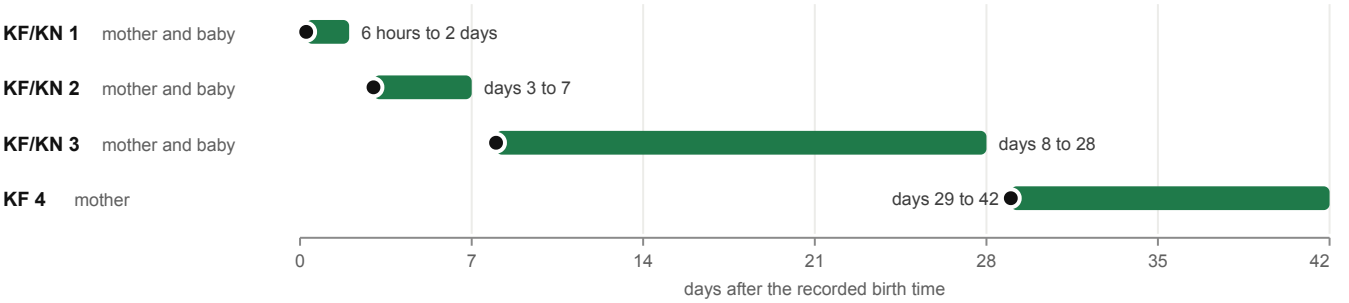


Figure 8. Recommended visit windows (`POSTPARTUM_VISIT_SCHEDULES` in index.js). The dot marks the earliest moment the reminder can go out: the first 2-hour window before her reminder time after the window opens.

Visit	What the reminder tells her to have checked
KF/KN 1	Mother: bleeding, infection, uterus, blood pressure. Baby: breathing, warmth, umbilical cord, early breastfeeding.
KF/KN 2	Mother: breastfeeding, postpartum danger signs. Baby: infection, jaundice, weight, feeding.
KF/KN 3	Mother: recovery, family-planning advice. Baby: growth, weight gain, exclusive breastfeeding.
KF 4	Mother: final postpartum check and choice of family-planning method.

Each visit is a text with its focus, then the poll *Apakah Ibu sudah melakukan kunjungan KF/KN 1?* (Have you had the KF/KN 1 visit?). **Belum** gets advice to see a health worker soon; **Sudah** gets thanks. At most one visit goes out per check, within the 3-a-day cap, so a birth recorded late does not flood her chat. **KF 4 = Sudah ends the program**: her status becomes completed and FE reminders stop.

Slow, spaced, and quick to stop

WhatsApp restricts numbers that behave like bulk senders. RemindCare runs on an ordinary number, so it sends like a careful person: one message at a time, with pauses, within daily limits, and it stops completely when something looks wrong. All of this lives in `lib/send-guard.js`.

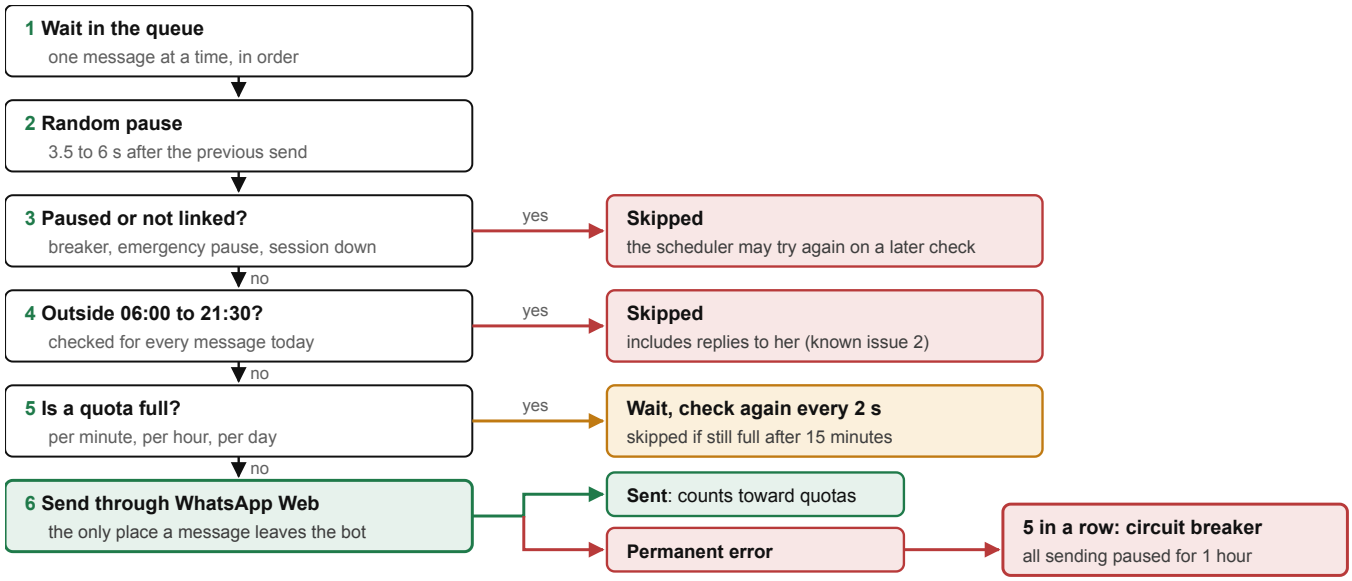


Figure 9. Every outgoing message, reply or reminder, takes this path. A permanent error means WhatsApp says the number is not registered, blocked or restricted; that mother is also paused. Other errors are temporary and retried later, as in Figure 11.

Daily cap while the session is new

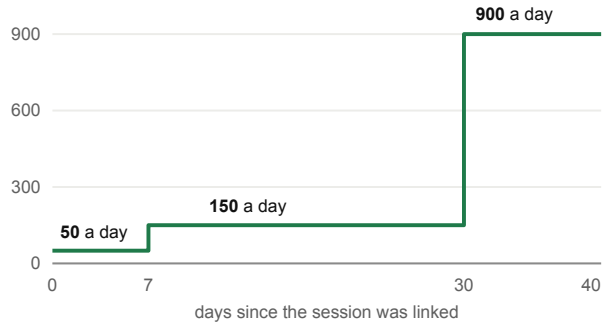


Figure 10. Brand-new sessions sending hundreds of messages are the most common way numbers get blocked. The date of first linking is kept in `data/wa_session_first_seen.txt`.

Waiting after a failed send

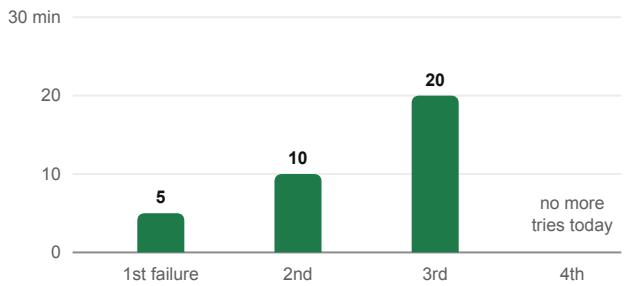


Figure 11. Minutes until the next try: 5 minutes doubling, capped at 30. Each reminder type gets 4 attempts per mother per day (`max_send_attempts`).

Limit	Default	Where to change it
Pause between sends	3.5 s plus up to 2.5 s random	Settings page, see page 7
Outgoing quotas	12 a minute, 180 an hour, 900 a day	Settings page
Warm-up	50 a day for a week, 150 until day 30	`SEND_WARMUP_*` in .env
Circuit breaker	5 permanent errors in a row, then 1 h pause	`SEND_FAIL_BREAKER`, `SEND_PAUSE_MS`
Per mother	3 scheduled sends a day	Settings page
Incoming messages	20 a minute per number, then 2 minutes ignored	`RATE_LIMIT_*` in .env

Also: the bot reports the version of the Chrome it really runs instead of the library's old default; only one copy may run per session; when WhatsApp drops or rejects the session the bot stops sending and raises an alarm instead of re-linking in a loop (after an authentication failure it pauses for 6 hours).

Follow every mother, change sending without a restart

The admin website only listens on the server itself (`127.0.0.1:3030`). Open it through an SSH tunnel, for example `ssh -L 3030:127.0.0.1:3030 your-server`, or put it behind a TLS reverse proxy.

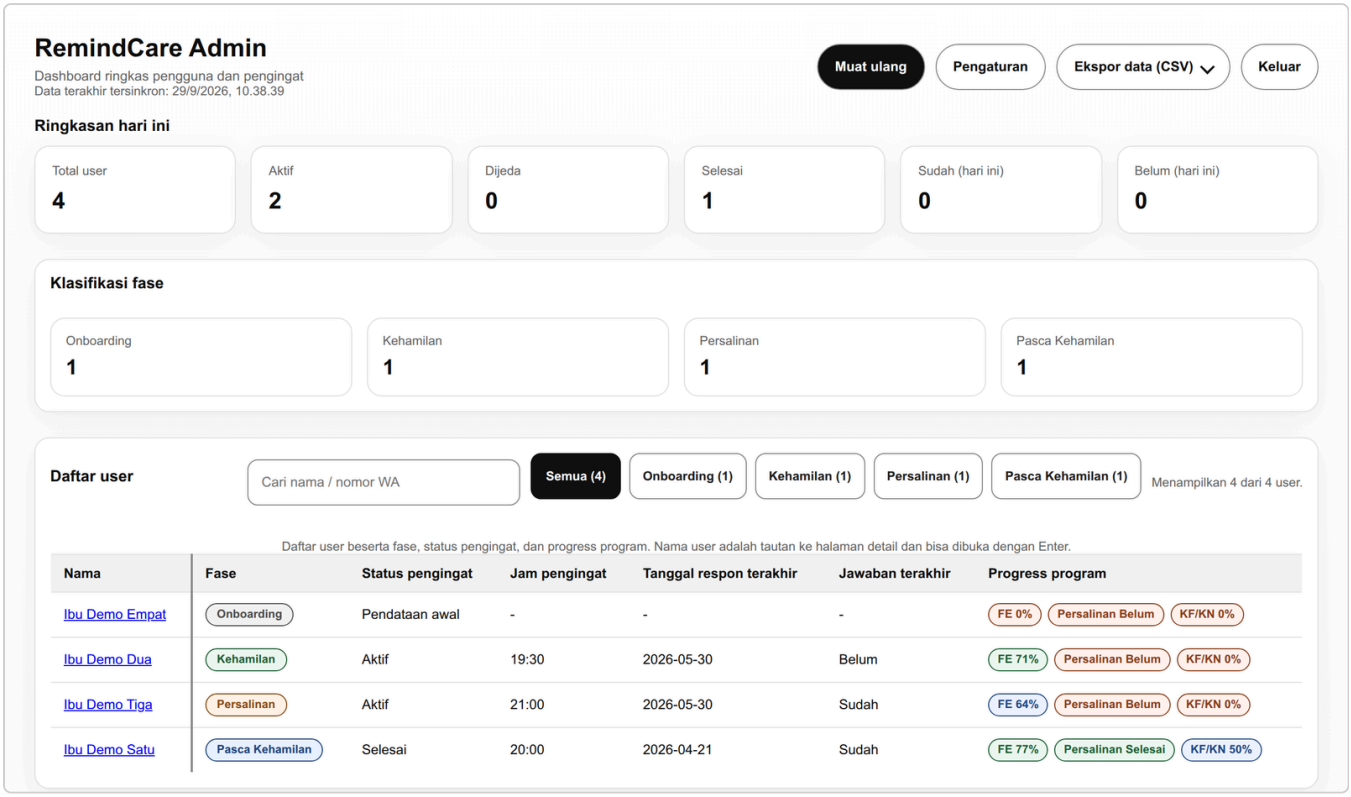


Figure 12. Dashboard (`/admin`) with the four demo mothers from `scripts/reset-seed-db.js`. Today's summary, mothers per phase, and the list with progress chips (FE %, birth, KF/KN %). CSV export covers all mothers, reminder logs or visit logs. A name opens her detail page: delivery data, visit and reminder history, and her own CSV.

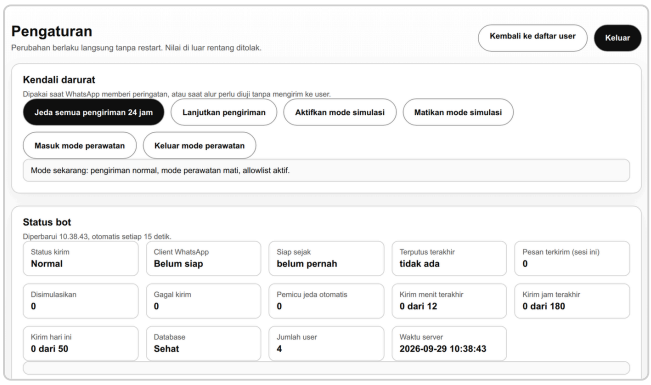


Figure 13. Settings (`/admin/settings`): emergency controls and live bot status, refreshed every 15 seconds. The limit fields follow below. "Kirim hari ini 0 dari 50" is the warm-up cap of a new session.

Controls

Emergency pause	Stops all sending for 1 to 168 hours (button: 24 h); survives a restart
Resume	Lifts the pause, resets the failure count
Maintenance	No scheduled reminders; the bot still answers
Dry run	Messages are written to the log, not sent
Limits	Range-checked, applied at once, no restart
Per mother	No buttons yet: `POST /admin/api/users/:waId/actions` with action `set_reminder_time`, `pause`, `resume`, `complete`, `clear_send_failures` or `delete_data` (value `HAPUS`)

Access

- User `admin`. Password from `ADMIN_WEB_PASSWORD`, or generated once into `data/admin_web_password.txt` (mode 600), stored as a scrypt hash.
- 5 failed logins from one IP block it for 15 minutes. A session lasts 8 hours (cookie `rc_admin`).

Admin commands in WhatsApp

For numbers in `ADMIN_WA_IDS`: **admin stats**, **admin allow**, **admin block** or **admin unblock** followed by a number, **admin purge logs** followed by days, and **admin help**.

Where the data lives and how to keep it safe

All program data sits in one SQLite file, and the WhatsApp session in a folder next to it. The database is copied every night; back up the session folder by hand once after linking.

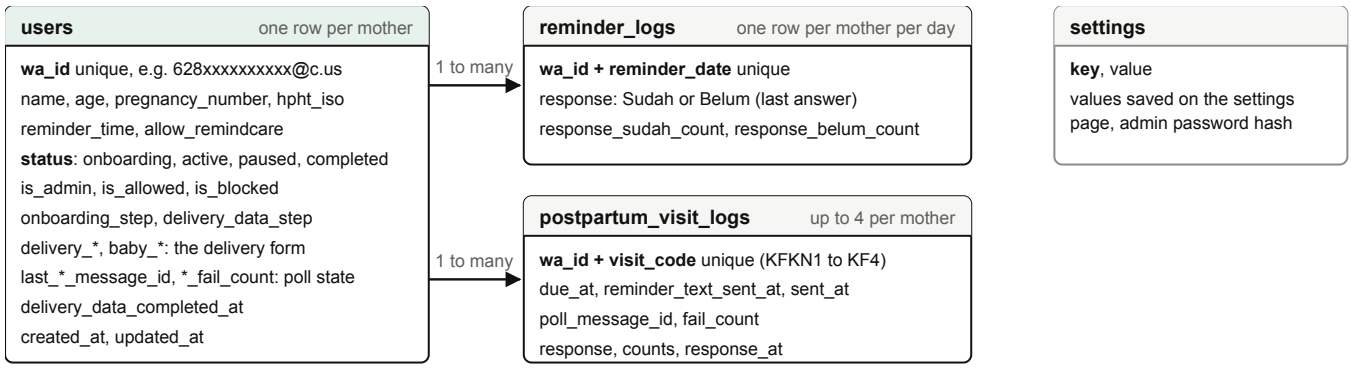


Figure 14. The four tables. Logs are linked to a mother by `wa_id` (no foreign key). Missing columns are added automatically at startup. Reminder logs older than 180 days are deleted once a day.

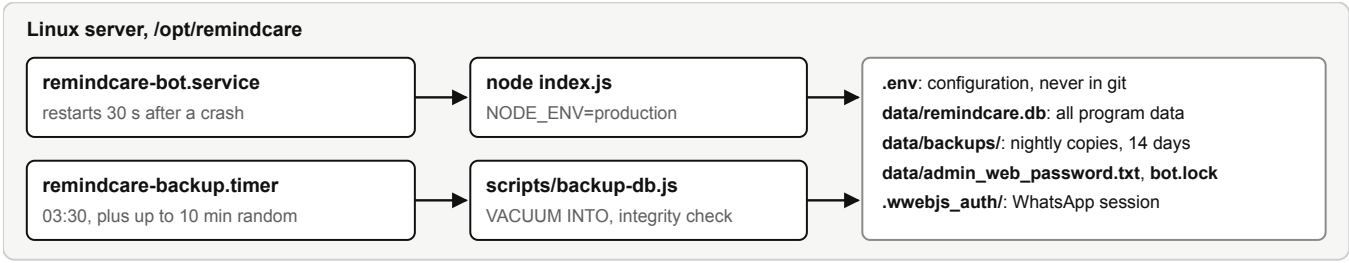


Figure 15. The units in `deploy/`. Enable the timer with `systemctl enable --now`; plain `enable` only arms it for the next boot. `data/` and `.wwebjs_auth/` are in `.gitignore`; check `git ls-files` before each commit.

Task	How (full steps in `deploy/INSTALL.md`)
Link the number	Scan the QR from `node index.js`, or set `WA_PAIRING_NUMBER` and enter the code in `data/pairing-code.txt`
Check it is live	`/admin/api/health` shows `"ready":true`; the journal shows *RemindCare siap digunakan*
Back up now	`systemctl start remindcare-backup.service`, then look for *Backup ok* in the journal
Restore the database	Stop, copy a file from `data/backups/` over `data/remindcare.db`, check integrity, start
Restore the session	Stop, replace `.wwebjs_auth/` with the archive made right after linking, start
New admin password	`node scripts/rotate-admin-password.js`; read it from `data/admin_web_password.txt`

Known issues in version 1.0

1. **"tidak mau" is read as yes.** `parseYesNo` checks yes-words first and *mau* matches, so declining question 8 with *tidak mau* or *ga mau* is saved as consent.
2. **Replies are treated like reminders.** `sendText` and `sendPoll` never pass `kind: "reply"`, so every reply waits 3.5 to 6 s and nothing is answered after 21:30. `REPLY_*` values are unused.
3. **Text-only days send the reminder twice.** `sendDailyPoll` sends the text before the poll check and again in the text-only branch.
4. **No end date after birth.** If KF 4 is never answered *Sudah*, FE reminders go on until an operator marks her completed.
5. **Some .env values are ignored.** At startup the settings-page values (or their defaults) replace `SEND_MIN_GAP_MS`, `SEND_MAX_PER_*`, `SEND_WINDOW_*` and `REMINDER_STALE_AFTER_MINUTES`. The retention field on the settings page is saved but not used.
6. **Reminder times outside 06:00 to 21:30** are accepted but never fire.